



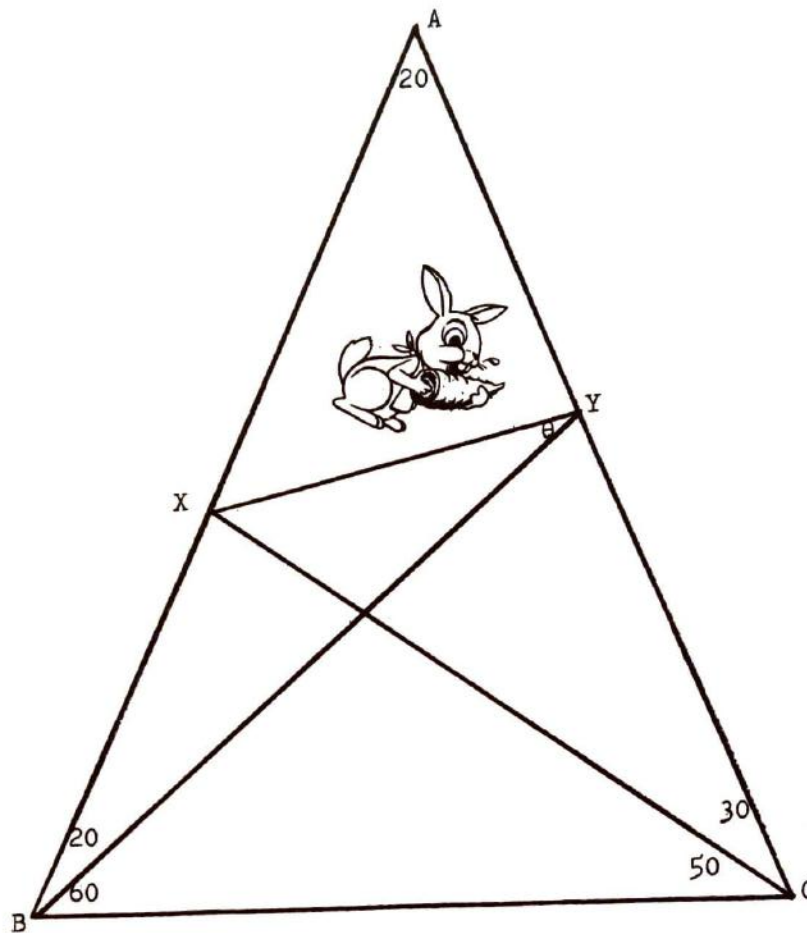
Bestseller?

Everything you
always wanted to
know about math*

Explained by
the **MATH** Spring, 1971
Stuyvesant
High School
SURVEY



PROBLEM



Given: Isosceles $\triangle ABC$ with vertex $\angle A = 20^\circ$, and $\angle XCB = 30^\circ$, $\angle YBC = 20^\circ$.

Find: $\angle BYX$

Hint on page 11

THE MATH SURVEY

PUBLISHED BY THE MATHEMATICS DEPARTMENT OF STUYVESANT HIGH SCHOOL

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The Math Survey

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Definitions

Congruences:

If the difference of two integers a and b is divisible by a third integer m , then we say a is congruent to b modulo m and write $a \equiv b \pmod{m}$. For example, $36 \equiv 2 \pmod{17}$, $14 \equiv 7 \pmod{13}$. Congruence, like equality, is an equivalence relation; that is, it satisfies the following three axioms:

(1) reflexivity - $a \equiv a \pmod{m}$

(2) symmetry - If $a \equiv b \pmod{m}$, then $b \equiv a \pmod{m}$

(3) transitivity- If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $a \equiv c \pmod{m}$

The first two axioms are obvious. A proof of (3) is as follows: If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then by the definition of congruence, $a-b = km$ and $b-c = jm$ for some integers k and j . Therefore, $(a-b) + (b-c) = a-c = (k+j)m$, and hence $a \equiv c \pmod{m}$.

The basic properties of congruences are listed below (unless otherwise specified, all congruences are assumed to be modulo m):

Thm. C a) If $a \equiv b$ and $c \equiv d$, then $a+b \equiv c+d$.

b) If $a \equiv b$ and $c \equiv d$, then $ac \equiv bd$.

Proof: let $a-b = km$, $c-d = jm$. Then $ac-bc = ckm$, and $bc-bd = bjm$. Therefore $(ac-bc) + (bc-bd) = ac-bd = (ck+bj)m$, so $ac \equiv bd$.

c) If $a \equiv b$, then $a^n \equiv b^n$.

Proof: let $a-b = km$. From the identity $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1}) = (a-b)X = kmX$, thus we see that $a^n \equiv b^n$.

d) If $f(x)$ is a polynomial $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n$ with integer coefficients, and $r \equiv s$, then $f(r) \equiv f(s)$.

Proof: This is easily proved using properties (a) and (c). The construction of a formal proof is left to the reader.

e) If $ax \equiv ay \pmod{m}$, then $x \equiv y \pmod{m/(a,m)}$, where (a,m) denotes the greatest common divisor of a and m (see def. of "greatest common divisor" in this section). Hence, if a is relatively prime to m (see def. of "relatively prime"), then $x \equiv y \pmod{m}$.

f) If $a \equiv b \pmod{m_i}$ for $i = 1, 2, 3, \dots, n$, then $a \equiv b \pmod{M}$, where M is the least common multiple of $[m_1, m_2, \dots, m_n]$ (see def. of "least common multiple").

g) (converse of (f)) If $a \equiv b \pmod{M}$ for $M = [m_1, m_2, \dots, m_n]$, then $a \equiv b \pmod{m_i}$ for $i = 1, 2, \dots, n$. In other words, if m is a divisor of M , then $a \equiv b \pmod{m}$.

For proofs of properties (e), (f), and (g), the reader can consult any number theory text, or if he is more ambitious, the reader may try to prove them himself (no simple matter).

II- Factorial ($n!$) - is defined by the recursive formula:

$$n! = n(n-1)!$$

In other words,

$$n! = n(n-1)(n-2)(n-3)\dots(3)(2)(1)$$

where $n \geq 0$ and $0! = 1$

III- Combination (${}_nC_r$) - A combination is a definite group of elements without any regard to their arrangement among themselves. From the n students in a class we want to select a committee of r ($n \geq r$). How many can be chosen?

Solution: We can choose the first member of our committee in any one of n ways, the second in $n-1$ ways, and so on. However, this result includes the various arrangements of these r students among themselves, which is $r!$. Since we are interested only in the number of selections we can make, we must eliminate the different arrangements within the group by division. Hence, the number of committees that can be chosen from the class is:

$${}_nC_r = n(n-1)(n-2)\dots(n-r+1)/r!$$

It can also be seen that ${}_nC_n = 1$, ${}_nC_1 = n$, and ${}_nC_r = {}_nC_{n-r}$

IV- Permutations (${}_nP_r$) - A permutation is an arrangement of a group of things in a definite order. Thus the three letters x, y, z can be arranged in six possible orders: $xyz, xzy, yxz, yzx, zxy, zyx$

Suppose we want to make arrangements from a group of n elements using r of them at a time. We have n choices for the first element, $n-1$ for the second, down to $(n-r+1)$. So we get:

$${}_nP_r = n(n-1)(n-2)\dots(n-r+1)$$

It may also be seen that ${}_nP_n = n!$ and ${}_nP_1 = n$

- V- Greatest Integer Function ($[]$)- For all real values of x , $[x]$ is the greatest integer which is less than or equal to x .
Examples: $[3/2] = 1$, $[-3/2] = -2$, $[1] = 1$
- VI- Prime- a number that has no factors except unity and itself.
- VII- Relatively Prime- Two numbers are said to be relatively prime when they have no common factors except unity. Two relatively prime numbers a and b are written: $(a,b)=1$
- VIII- Prime factorization- Any number can be factored into the form:
$$N = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$$
where k_i is an integer, and p_i is a prime for $i=1,2,3,\dots,r$.
- IX- Aliquot divisors- all divisors, including unity, of any given number.
- X- Perfect Numbers- a number which is equal to the sum of its aliquot divisors.
Examples: 6 and 28. ($6 = 3+2+1$, $28 = 14+7+4+2+1$)
- XI- Iff ($\Leftrightarrow$)-means if and only if.
- XII- $a|b$ - means that a is a factor of b .
- XIII- Gamma Function (Γ)- is defined for positive integers n as $(n-1)!$
- XIV- Euler's ϕ (phi) function- for any integer N ,
$$N = p_1^{k_1} p_2^{k_2} p_3^{k_3} \dots p_r^{k_r}$$
$$\phi(N) \text{ is defined as the number of integers less than } N \text{ and prime to it, and}$$
$$\phi(N) = p_1^{k_1-1}(p_1-1)p_2^{k_2-1}(p_2-1)\dots p_r^{k_r-1}(p_r-1)$$
- XV- Greatest Common Divisor- the largest positive common divisor of two integers a and b
- XV- Greatest Common Divisor- If a and b are nonzero integers, the greatest common divisor (g.c.d.) of a and b is the unique positive integer d with the following two properties:
1- $d|a$ and $d|b$.
2- If c is an integer such that $c|a$ and $c|b$, then $c|d$.
- XVI- Least Common Multiple- The least common multiple (l.c.m.) of two nonzero integers a and b is the positive integer m with the following two properties:
1- $a|m$ and $b|m$
2- If c is an integer such that $a|c$ and $b|c$, then $m|c$.

!!! REWARD !!!

The Math Survey will reward the first five students who can solve any of the following problems. To claim the rewards (each reward is listed next to its respective problem), bring all solutions to the Math Office (Room 312A).
Deadline: 1984. (Sorry, this contest is not open to teachers.)

1. Prove that the equation $x^n + y^n = z^n$ has no positive integral solution for (x,y,z) where n is an integer greater than 2. (Reward- \$5.00)
2. Prove that every even number may be expressed as the sum of two primes. (Reward- \$2.50)
3. Find a general formula in x which will yield the values of x for which $2^x - 1$ is a prime. (Reward- \$4.00)
4. Archimedes' Cattle Problem: Determine from the following conditions the number of cattle that once grazed on an island off Italy. The cattle were divided into four herds- white, black, dappled, and yellow, and the number of bulls was greater than the number of cows. The conditions:
 - a. The number of white bulls = the number of yellow bulls plus $1/2 + 1/3$ the number of black bulls.
 - b. The number of black bulls = the number of yellow bulls plus $(1/4 + 1/5)$ the number of dappled bulls.
 - c. The number of dappled bulls = the number of yellow bulls plus $(1/6 + 1/7)$ the number of white bulls.
 - d. The number of white cows = $(1/3 + 1/4)$ the number in the black herd.
 - e. The number of black cows = $(1/4 + 1/5)$ the number in the dappled herd.
 - f. The number of dappled cows = $(1/5 + 1/6)$ the number in the yellow herd.
 - g. The number of yellow cows = $(1/6 + 1/7)$ the number in the white herd.
 - h. The number of white bulls plus the number of black bulls = a square number.
 - i. The number of dappled plus yellow bulls = a triangular number (i.e. a number in the form $n(n+1)/2$). (Reward \$5.00—handwritten solutions only)
5. Present a trisection of an angle in the general case. (Reward- \$3.00)
6. John starts at a certain time driving his car from New York to Washington traveling at a rate of 100 miles per hour. Sixty minutes later, Martha leaves in her car enroute from Washington to New York, doing 75 miles in an hour. The distance between New York and Washington is $\pi^{4.77}$. When the two cars meet, which one is nearer to New York? (Reward- Free Errata sheet)



Ambiguities in Probability

by Malcolm Cheung

What is the probability that a chord drawn at random within a given circle will be longer than the side of an equilateral triangle inscribed in that circle?

There are three ways of approaching this problem. All three are valid. All three lead to different answers.

Method 1: The random chord drawn will have a midpoint that can land anywhere within the circle. From Figure 1, we can see that all the chords tangent to the circle inscribed in the triangle (i.e. whose midpoints lie on the circumference of the inscribed circle) will be equal in length to the side of the equilateral triangle. It can also be seen that all the chords whose midpoints lie within the inscribed circle will be longer than the side of the triangle. The radius of the inscribed circle is equal to one half the radius of the given circle. Therefore its area is $1/4$ that of the given circle and the probability that a chord's midpoint will land in the region is $1/4$.

Answer by Method 1: $1/4$

Method 2: The chord drawn will originate at a random point on the circumference of the circle. Once this point is determined, the other endpoint will be any other point on the circumference. From Figure 2, it can be seen that only those endpoints between vertices A and B of inscribed equilateral triangle ABC will result in chords longer than the side of the triangle. Since angle C is 60° , arc AB is 120° or $1/3$ the total circumference. Therefore, the probability that the second endpoint will land on arc AB is $1/3$.

Answer by Method 2: $1/3$

Method 3: Any chord drawn is perpendicular to one of the circle's diameters (See Figure 3). All chords perpendicular to this diameter (whichever we pick) pass through the diameter at points that range uniformly along it. It can be seen in Figure 3 that only those chords falling between sides AB and DE, of inscribed equilateral triangles ABC and DEF respectively, will be longer than those sides. It can be proven that OA' and OD' each equal $1/2$ the radius of the circle. Therefore the distance between AB and DE is equal to the radius or $1/2$ the diameter of the circle and the probability that the chord drawn will pass through diameter CF between points A' and D' is $1/2$.

Answer by Method 3: $1/2$



Figure 1

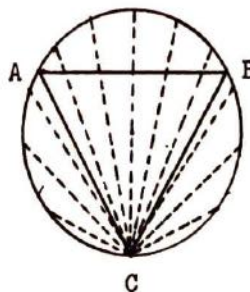


Figure 2

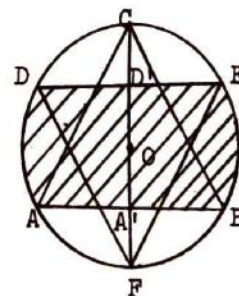


Figure 3

Such a problem can be described as "ambiguous" or "double-tongued" or even, as we have just seen, "triple-tongued." The primary defect in the so-called "ambiguous" problems is the failure to specify the randomizing procedure. Again, all the above results are acceptable. We can arrive at a definite answer to the problem if and only if the method of drawing a chord at random is specified. Of course, in actual experimentation, none of the above methods are utilized by the subject. We just draw any chord that happens to strike our fancy. It is interesting to note that, according to Dr. Oliver L. Lacey, professor of psychology at the University of Alabama, far more than 50 per cent of his subjects actually drew a chord longer than the side of an inscribed equilateral triangle; a result that doesn't agree with any of the previous answers.

Consider the following problem: Mr. Smith has two children. At least one of the two is a boy. What's the probability that both his children are boys?

In the above form, this question is unanswerable for there are two procedures for arriving at two different answers. First, if a family is chosen at random from the set of all families with two children, at least one of whom is a boy, the answer is $1/3$.

Boy-Boy

Boy-Girl
Three Equally Probable Cases

Girl-Boy

On the other hand, if a family is chosen at random from the set of all families with two children, regardless of sex, the answer is $1/4$.

Boy-Boy

Girl-Girl

Boy-Girl

Girl-Boy

Four Equally Probable Cases

This ambiguity can be remedied if the problem were worded as follows: Mr. Smith's family has two children of whom at least one is a boy. What is the probability that the second child is a boy? In this case, it is quite clear that we are dealing only with those families with two children of whom at least one is a boy and that the answer is $1/3$.

However, many problems which are not ambiguous appear so in the mind of the person who is trying to solve them. For instance: Three cards, two green and one black, are dealt, face down, to three people—A, B and C. What is the probability that A has the black card? The obvious answer is $1/3$. Now suppose C's card is turned over and is found to be green. Again, what is the probability that A has the black card? This time, because, of the two remaining cards, one is black, one is inclined to answer $1/2$. This reasoning is wrong. No matter what the color of A's card, either B or C must have a green card. Therefore, revealing C's card as green provides no additional information that can be of use and the probability remains $1/3$.

Another difficulty arises in problems dealing with an infinite number of possibilities. These problems are not really ambiguous and not quite paradoxical. What follows are two problems which lead to a startling although invalid conclusion.

A real number, rational or irrational, between 0 and 10 is chosen at random. What's the probability that the number picked is greater than 5?

Looking at the number line (Figure 4), the answer is $1/2$.

A real number, rational or irrational, between 0 and 100 is chosen at random. What's the probability that the number picked is greater than 25?

Again, the number line (Figure 5) suggests an answer of $3/4$.

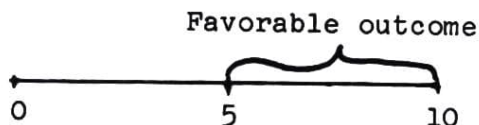


Figure 4

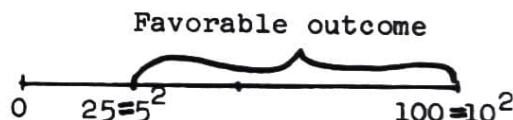


Figure 5

Since every number between 25 and 100 has a square root between 5 and 10, we can come to the conclusion that in selecting a number at random between 0 and 10 the probability that it is greater than 5 is $1/2$ while the probability that it's greater than 5 if the square of the number is chosen at random is $3/4$!

This problem is cleared up once we accept the fact that there are just as many real numbers between 0 and 25 as there are between 25 and 100. However, the most rational explanation of the problem was that of Bertrand. In essence, what he said was that one cannot express choices made at random from an infinite number of possibilities in terms of finite probabilities.

The history of probability theory has seen bitter controversies over even some of the most basic concepts. They arise over ambiguities and paradoxes and go unsettled for years. Some may never be settled. Perhaps the best attitude for one to take is the pragmatic one or, in the words of Aristotle, "The probable is what usually happens." But mathematicians are not pragmatists

and they would probably agree with Euripides when he said, "There is safety in numbers."

Euler's Theorem

by Robert Berman

What is the smallest number such that moving the right hand digit to the front of the number (and shifting all the other digits one place to the right) doubles the number? In solving this problem, two notions must be used.

(1) modulus

(2) Euler's ϕ (phi) function

(See definition section for a basic explanation of them.)
First it is important to mention several theorems involving Euler's ϕ function.

(a) $\phi(p) = p-1$ (where p is a prime)

(b) $\phi(p^n) = p^{n-1}(p-1)$ (where p is a prime)

(c) $\phi(a)\phi(b) = \phi(ab)$ (where a and b are relatively prime)

(d) If m is a positive integer, $m = p_1^{n_1} p_2^{n_2} p_3^{n_3} \dots p_r^{n_r}$, where p_x is a positive prime, and n_x is a positive integer for $x=1, 2, \dots, r$,
 $\phi(m) = p_1^{n_1-1} p_2^{n_2-1} \dots p_r^{n_r-1} (p_1-1)(p_2-1) \dots (p_r-1)$

(For proofs of all these theorems see Math Survey, Volume XLIV, (Spring 1969), Euler's Phi Function, by A. Rothstein.)

Understanding how to arrive at the exact value of $\phi(m)$, where m is any positive integer, is essential to this problem so we will give special attention to theorem (d).

Since every number m can be broken down into prime factors, we can see that some of the above theorems may be employed in determining the exact value of $\phi(m)$.

From theorem (c)

$$\phi(m) = \phi(p_1^{n_1}) \phi(p_2^{n_2}) \dots \phi(p_r^{n_r})$$

From theorem (b)

$$\phi(m) = p_1^{n_1-1} (p_1-1) p_2^{n_2-1} (p_2-1) \dots p_r^{n_r-1} (p_r-1)$$

This is equivalent to the theorem and proves it.

Euler's theorem states:

$$a^{\phi(N)} \equiv 1 \pmod{N} \quad (\text{where } a \text{ is prime to } N)$$

Examples:

$$a^{\phi(3)} \equiv 1 \pmod{3} \text{ is equivalent to } a^2 \equiv 1 \pmod{3}$$

This states that the square of any number that is not a multiple of 3 has a remainder of 1 when divided by 3. (E.g. $5^2 = 25 \equiv 1 \pmod{3}$)

In the same way substituting 4 for N we get that the square of any number that is not a multiple of 4 has a remainder of 1 when divided by 4. (E.g. $3^2 = 9 \equiv 1 \pmod{4}$)

Proof of Euler's Theorem:

$$a^{\phi(N)} \equiv 1 \pmod{N} \text{ where } a \text{ is prime to } N.$$

Let the solutions of the Euler $\phi(N)$ function be:

$$(s_1, s_2, s_3, \dots, s_{\phi(N)})$$

The solutions are said to form a reduced residue system mod N . Multiply each of these by a , and represent that product as an integral number of N 's plus a remainder.

$$as_1 = Q_1 N + R_1; \quad as_2 = Q_2 N + R_2; \quad \dots \quad as_{\phi(N)} = Q_{\phi(N)} N + R_{\phi(N)}$$

In every case it may be seen that each R is different from every other R . If any 2 R 's, say R_x and R_y were equal, we could subtract the two equations and get:

$$as_x - as_y = (Q_x N - Q_y N) + R_x - R_y \quad \text{and since } R_x = R_y$$

$$a(s_x - s_y) = N(Q_x - Q_y)$$

In order for this equation to be valid, N must be a factor of the left side, but all s 's are smaller than N and prime to it, and a is prime to N . This contradiction shows that the last equation is impossible.

Since $R_1, R_2, \dots, R_{\phi(N)}$ are all distinct and form a residue system of $\phi(N)$ numbers, each R_j will equal some s_i . We can now multiply the equations together:

$$(as_1)(as_2) \dots (as_{\phi(N)}) = (Q_1 N + R_1)(Q_2 N + R_2) \dots (Q_{\phi(N)} N + R_{\phi(N)})$$

$$= a^{\phi(N)} (s_1 s_2 \dots s_{\phi(N)}) = N(Q_1 Q_2 \dots Q_{\phi(N)} + \text{cross terms of } R\text{'s and } Q\text{'s}) + R_1 R_2 \dots R_{\phi(N)}$$

OPTICAL ILLUSION
Which is larger?



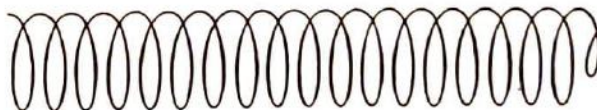
Answer: The one on the right.

$$\begin{aligned} a^{j(N)} s_1 s_2 \dots s_j \phi(N) - s_1 s_2 \dots s_j \phi(N) &= \text{IN} \\ (a^{j(N)} - 1)(s_1 s_2 \dots s_j \phi(N)) &= \text{IN} \end{aligned}$$
$$a^{\phi(N)} - 1 \equiv 0 \pmod{N} \quad \text{or} \quad a^{\phi(N)} \equiv 1 \pmod{N}$$
$$N = a_k 10^k + a_{k-1} 10^{k-1} + \dots + a_1 10^1 + a_0 10^0 \quad \text{and}$$
$$20N = a_0 10^{k+1} + a_k 10^k + \dots + a_2 10^2 + a_1 10^1$$
$$19N = a_0[10^{k+1} - 1] \quad \text{and} \quad N = a_0[10^{k+1} - 1]/19$$

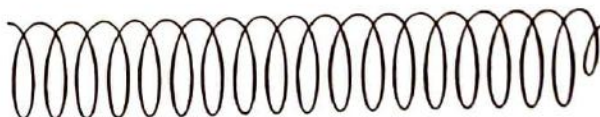
$(10^{18}-1)/19$ which is also known as 052631578947368421.

$$N = (10^{k+1} - 1)/29$$

99999999999999999999999999999999/29 equals.



Once a young lady called Lillian
Summed the numbers from one to a billion.
But it gave her the fidgets
To add up the digits;
If you can help her, she'll thank you a million.



9

Groups

by John Chin

A mathematical system consists of a set of objects, one or more operations on these objects, and certain assumptions about the objects, and the operations. One of the simplest, and also one of the most important, mathematical systems is the group. A group has only one operation, $\circ$, which may be "addition," or may be "multiplication," or may be the operation "following" or "combining," or may be any other operation. (The reader will note that operation is an undefined term.)

Definition: A group G is a nonempty set of elements and a binary operation $\circ$ for which the following conditions are satisfied.

A. G is closed with respect to $\circ$.

B. The associative law holds in G ; that is

$$a \circ (b \circ c) = (a \circ b) \circ c, \text{ when } a, b, \text{ and } c \text{ are elements in } G.$$

C. There is an identity e in G such that

$$e \circ a = a = a \circ e, \text{ for all elements } a \text{ in } G.$$

D. For each element a in G there is in G an inverse element of a denoted by a' (read "a inverse") such that

$$a \circ a' = e = a' \circ a.$$

Two elements a and b are said to commute with respect to an operation $\circ$ if $a \circ b = b \circ a$. The center of a group is the set of all elements of the group which commute with every element of the group. Every group has a center: the identity element commutes with every element of the group, therefore it must be in the center. For some groups the center consists of just the identity; for others it contains more than just the identity, yet does not contain all the elements, and is therefore a proper subset of the set of elements in the group. If the center is the whole group, the group is called a commutative or abelian group.

We, being students of higher mathematics, might wonder how many elements of a group behave like an identity. If an element of a group leaves every element of the group unchanged when operating on the left, we call it a left identity. (The definition of a right identity is left as an exercise for the reader.)

We now consider the relationship between a left or right identity and the identity of the group. By common sense, we assume correctly that the identities are not distinct but are equal, and therefore, that the identity is unique. One of the proofs is as follows:

Discussion

Reason

Assume that there are two identities-- e and f . Indirect argument

Then $e \circ f = f$.

By condition C, since e is an identity.

Also $e \circ f = e$.

Also by condition C.

Therefore $e = e \circ f = f$

The operation is well-defined

or $e = f$

and there is only one identity.

The question of the number of inverses that each element of a group has arises. To answer it, let us assume that an element of a group, denoted by a , has its usual inverse a' and another inverse which we shall denote by y . Since a and y both satisfy condition D we have

$$a \circ a' = e = a' \circ a$$

and

$$a \circ y = e = y \circ a.$$

Therefore

$$a \circ a' = a \circ y.$$

Let a' operate on the left of each side of this equality. This produces $a' \circ (a \circ a') = a' \circ (a \circ y)$. By applying the associative law we have $(a' \circ a) \circ a' = (a' \circ a) \circ y$. By using the associative law again we have $e \circ a' = e \circ y$. Using condition C we have $a' = y$. Therefore, every element that acts like an inverse is a' , and a has only one inverse.

A nonempty subset H of a group G is a subgroup of G if H is a group with respect to the operation of G . Every group has at least two subgroups--the whole group and the group consisting of the identity alone. These two subgroups are called improper subgroups, and all other subgroups are called proper subgroups.

Example- The set of all even integers is a subgroup of the set of all integers under addition. In fact, the set of all integral multiples of a fixed integer k is a subgroup of the integers under addition.

A finite group is generated by the elements $a_1, a_2, a_3, \dots, a_n$ if every element of the group can be expressed in terms of these elements and the group operation. When a group G can be generated by an element h in G , G is a cyclic group. The order of a group element is the order of the cyclic subgroup which it generates.

To find the subgroups of a given finite group systematically, first generate subgroups using each of the elements, then use combinations of elements taken twice, and so until it is no longer possible to increase the number of group generators.

Let S be a subgroup of a finite group G . We can consider a certain set of group elements of the form $s \circ r$ where s is an arbitrary element in S and r is a specified element in G . This set is a right coset of S , denoted by S_r . A left coset if, appropriately, a set of the form $r \circ s$, but we shall concern ourselves only with right cosets since all statements which hold true for right cosets will hold true for left cosets.

A right coset of a subgroup S of G has the same number of elements as S . For each s in S there is a unique element $s \circ r$ in S_r . Therefore, S_r does not have more elements than S . If two elements $s_1 \circ r$ and $s_2 \circ r$ of S_r are equal, then $s_1 \circ r = s_2 \circ r$. Since r^{-1} exists in G then $(s_1 \circ r) \circ r^{-1} = (s_2 \circ r) \circ r^{-1}$, $s_1 \circ (r \circ r^{-1}) = s_2 \circ (r \circ r^{-1})$, $s_1 \circ e = s_2 \circ e$, and $s_1 = s_2$. Therefore distinct elements in S correspond to distinct elements in S_r , and S_r cannot have fewer elements than S ; therefore, S and S_r have the same number of elements.

If S_h and S_k are two right cosets of the same subgroup S , then either $S_h = S_k$ or the two cosets have no elements in common. If there is an element in both S_h and S_k , then there are elements s_1 and s_2 in S such that $s_1 \circ h = s_2 \circ k$. Since S is a subgroup, s_2^{-1} exists in S , and

$$s_2^{-1} \circ (s_1 \circ h) = s_2^{-1} \circ (s_2 \circ k).$$

$$\text{Then } (s_2^{-1} \circ s_1) \circ h = (s_2^{-1} \circ s_2) \circ k \\ = e \circ k = k,$$

$$\text{and for every } s \text{ in } S, \quad s \circ [(s_2^{-1} \circ s_1) \circ h] = s \circ k, \\ [s \circ (s_2^{-1} \circ s_1)] \circ h = s \circ k.$$

Since $s \circ (s_2^{-1} \circ s_1)$ is an element of S , and every element of S_k can be expressed in the form $s \circ k$, where s is in S , then every element in S_k is in S_h .

A similar argument shows that every element of S_h is in S_k and therefore $S_h = S_k$ if the cosets have even one element in common; and if the cosets do not have one element in common then they have no elements in common.

Lagrange's Theorem: The order of a finite group G is an integral multiple of the order of each of its subgroups.

Discussion: Let S be a subgroup of G , then G can be divided into distinct right cosets each of which has the same order as S . If $S = \{j, k, l\}$ and there is an element h of group G but not in S then the elements of right coset S ($\{j \circ h, k \circ h, l \circ h\}$) are not equal to the elements of S by the above statement. Since the elements of S are distinct elements of the group, there must be at least six elements in G , then three of them are in S and three of them are in S and three are in S . If there is another element i which is neither in S or S , the order of G increases again by three and so on. By continuing this, we can see that the finite group G can be separated into cosets which have no elements in common, and the order of G must be an integral multiple of the order of S , in this case, three.

Examples of groups:

I. Cut out a square of cardboard or paper. Label the vertices A, B, C, D as shown below:



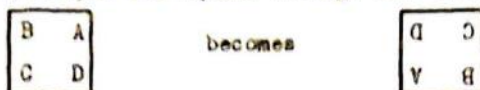
Place a letter A on the back face of the square in the same vertex as the letter A on the front. Vertex A is now uniquely identified. Repeat for B, C , and D .

The elements of this group are certain movements of the square. The operation, $\circ$, consists of following one movement by another. Equality is identity of vertex positions.

~~~~~

Hint for geometry problem on inside cover: \_\_\_\_\_ Complete solution is found on page 31.  
Draw  $BZ$  so that  $m\angle ZBC = 20^\circ$  and join  $XZ$ .

The permitted movements (elements) are:  
 $R$ : A rotation (in the plane) of the square through  $180^\circ$  about its center.



$R^\circ$ : A rotation (in the plane) of the square through  $360^\circ$ .



$R$  is the identity element, since the positions of the vertices are unchanged.  
 $H$ : A flip (in 3-space) about a horizontal line through the center of the square.



$V$ : A flip (in 3-space) about a vertical line through the center of the square.



Then  $(H \circ V)$  is that element which results when  $H$  (horizontal flip) is followed by  $V$  (vertical flip). Actual experiment will show that this is a group. A operation table will evince this:

| $\circ$   | $R^\circ$ | $R$       | $H$       | $V$       |
|-----------|-----------|-----------|-----------|-----------|
| $R^\circ$ | $R^\circ$ | $R$       | $H$       | $V$       |
| $R$       | $R$       | $R^\circ$ | $V$       | $H$       |
| $H$       | $H$       | $V$       | $R^\circ$ | $R$       |
| $V$       | $V$       | $H$       | $R$       | $R^\circ$ |

A. Closure: Only the elements  $R$ ,  $R^\circ$ ,  $H$ ,  $V$  are needed to complete the table.

B. Associative Law: The four given elements are substitutions of the corner letters. Substitutions are associative (the proof of this will not be given here, but can be found in Selections from Modern Abstract Algebra by Richard V. Andree).

C. Identity:  $R^\circ$

D. Inverses: Since  $e = R^\circ$  appears once, and only once, in each row and column of the operation table, it follows that  $x \circ a = R^\circ$  and  $b \circ x = R^\circ$  are solvable for all  $a, b$  in the group. Examination shows that, if  $x \circ y = R^\circ$ , then  $y \circ x = R^\circ$ .

II. A set of permutations can also form a group; the following set of permutations form a finite, nonabelian, cyclic group:

$$e = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} \quad a = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

The operation,  $\circ$ , consists of performing one permutation after the other.

Example-  $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} \circ \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix} = e$

The operation table is as follows:

| $\circ$ | $e$ | $a$ | $b$ |
|---------|-----|-----|-----|
| $e$     | $e$ | $a$ | $b$ |
| $a$     | $a$ | $b$ | $e$ |
| $b$     | $b$ | $e$ | $a$ |

A. Closure: Only the elements  $e, a, b$  are needed to complete the table.  
 B. Associative Law: Permutations are associative (for proof see the above references).  
 C. Identity:  $e$   
 D. Inverses:  $e o e = e = e o e$   
 and  $a o b = e = b o a$   
 therefore  $a' = b$   
 $b' = a$   
 $e' = e$ .

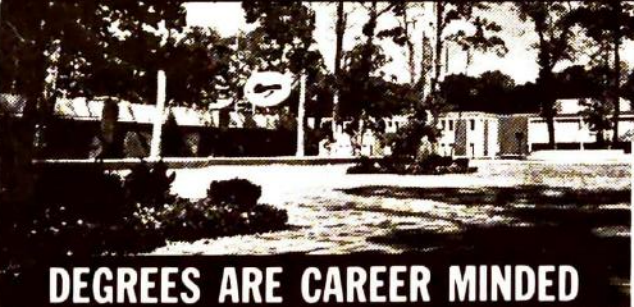
The group generators of this cyclic group are  $a$  and  $b$ . The positive powers of both  $a$  and  $b$  yield the elements of the permutation group:

$a^1 = a$   
 $a^2 = a o a = b$   
 $a^3 = a o a o a = e$   
 and  $b^1 = b$   
 $b^2 = b o b = a$   
 $b^3 = b o b o b = e$ .

Note: It can be seen here that if  $x$  is a generator of a cyclic group, and  $n$  is the order of the group, that  $x^n$  is equal to the identity of the group. An interested reader may wish to probe into it further.



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# The Trisection of an Angle

by Douglas Fried

Most Stuyvesantians have found ways to bisect any angle or any line segment; they have even managed to find ways to trisect any line segment. A few self-proclaimed wizards have even found ways to trisect any given angle—or so they say. There are several simple ways to trisect an angle, the easiest of which is to use a protractor. When dealing with constructions, however, mathematicians make life difficult. They allow the use of only an unmarked straightedge and a compass, and they require that the construction have a finite number of steps. If you disregard either of these rules, it becomes fairly easy to trisect the angle.



Fig. 1

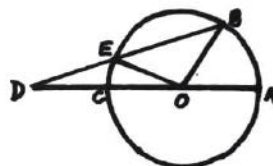


Fig. 2

In figure 1,  $\angle AOB$ , the given angle, is bisected, then  $\angle BOB'$  is bisected, then  $\angle B'O''B''$  is bisected, etc.. After an infinite number of steps,  $\angle AOB$  will have been trisected, since an angle is obtained which is equal to  $(1/2 - 1/4 + 1/8 - 1/16 + 1/32 - \dots)$  multiplied by the measure of the original angle. This infinite series when summed up by using the formula  $S = a/(1-r)$ , is equal to  $1/3$ . Another "illegal" method involves using the straightedge to measure distances. In Figure 2, a circle with any radius is constructed around the given  $\angle AOB$ . Diameter  $AC$  is extended to  $D$  such that  $DE$  is equal to the radius of the circle. Since exterior angles are equal to the sum of the opposite two interior angles of a triangle, and since base angles of an isosceles triangle are equal,  $\angle AOB \approx \angle OBD + \angle EDO \approx \angle EDO + 2\angle EDO \approx 3\angle EDO$ . Q.E.D.

With the original conditions, the trisection of any given angle can be proven to be impossible. In the proof, an important fact must be noted, namely that only those values which can be obtained from a finite number of rational operations or the extractions of square roots, can be constructed. (e.g.  $\sqrt[3]{2}$  is non-constructable whereas  $1+\sqrt{2}$  can be drawn. The proof is as follows:

$$\begin{aligned} (\cos \theta - i \sin \theta)^3 &= \cos(3\theta) - i \sin(3\theta) \quad (\text{DeMoivre's Theorem}) \\ \text{also, } (\cos \theta - i \sin \theta)^3 &= \cos^3 \theta - 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \\ \text{by equating real parts, } \cos(3\theta) &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta \\ &= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) \\ &= 4 \cos^3 \theta - 3 \cos \theta \end{aligned}$$

Now, let  $\theta = 20^\circ$  and  $\cos \theta = x/2$

$$\text{Therefore, } 1/2 = x^3/2 - 3x/2 \quad \text{or, } x^3 - 3x - 1 = 0 \quad (1)$$

The above equation, by the rational root theorem, can have only  $\pm 1$  as its only possible rational roots. However neither 1 nor -1 are roots of the equation. Now all that remains is to eliminate any possible irrational values. (Imaginary values are non-constructable.) Since in an equation with real coefficients irrational roots appear in conjugate pairs, if any irrational roots exist, they must be in the form of  $a+b\sqrt{2}$  and  $a-b\sqrt{2}$ . Also, the sum of the roots of eq. (1) is equal to zero since the sum of the roots of any  $n$ th degree equation of the form  $a_0x^n + a_1x^{n-1} + \dots + a_n = 0$  is equal to  $-a_1/a_0$ . A third degree equation may have at most three roots and thus a third root must exist with the value  $-2a$  (a rational value) in order for the sum of the roots of (1) to be zero. Consequently, eq. (1) has no real solution, or in other words,  $\cos 20^\circ$  can not be constructed. Finally, it can easily be seen that an angle can be constructed iff its cosine is constructable. We have now shown that one can not construct an angle of  $20^\circ$  and thus one can not trisect an angle of  $60^\circ$  (or any general angle.)

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# Diophantine Analysis

by Jerry Schonfeld

The theory of Diophantine analysis concerns itself with two closely related but somewhat different problems. The first problem is to find rational solutions of  $f(x,y,z,\dots)=0$ , which is a polynomial with rational coefficients. We usually make further restrictions on the problem by requiring that the solutions shall consist of integers. Similarly, if we have several such equations, the amount of which is less than the number of variables, then the set of equations is said to be a Diophantine system of equations.

A certain extension of the above is possible. One may replace  $f(x,y,z,\dots)$  by another which is not necessarily a polynomial. Take as an example all positive integral solutions of  $x^y - y^x = 0$ . Set  $y = tx$ , where  $t$  is any rational number. Thus,  $x^{tx} = (tx)^x$  or  $x = t(1/t-1)$ ,  $y = t(t/t-1)$ . For  $x$  and  $y$  to be integers,  $x^y$ , observe that for  $t > 2$ ,  $y$  is never an integer and so  $x=2$ ,  $y=4$ , (or  $(2,4)$ ), is the only integral solution. To obtain rational solutions set  $x = (a+1/a)^2$ ,  $y = (a+1/a)^{2+1}$  which satisfies the above condition. Therefore rational solutions are included in the set  $\{(3^2/2^2, 3^3/2^3), (4^3/3^3, 4^4/3^4), \text{etc.}\}$

A particular solution to a Diophantine problem may be obtained by different methods. The difficulty lies in proving that your solution gives the general solution. For instance, take the equation  $x^4 + y^4 = z^4$ .

Leonhard Euler proved that  $a^4 + b^4 \neq *$  (where  $*$  denotes a square whose exact value is not needed in the proof) for all positive integers  $a, b$ . He proceeded thus: if  $(a^2)^2 + (b^2)^2 = *$ , where  $a$  and  $b$  are relatively prime, then  $a^2 = p^2 - q^2$ ,  $b^2 = 2pq$  where  $p$  and  $q$  are of a different parity (i.e. one is odd and the other even). Let  $p$  be odd and  $q$  be even. Then  $(p)(2q) = b^2$  where  $p$  and  $2q$  are relatively prime and therefore both squares. From  $p^2 = a^2 + q^2$ , we get  $p = m^2 + n^2$  and  $q = 2mn$ , where  $m$  and  $n$  are relatively prime and of different parity. Since  $2q = *$ ,  $mn = *$  and  $m = x^2$ ,  $n = y^2$ . Therefore,  $x^4 + y^4 = p = *$ , where  $x < 0$  and  $y < b$ . Since we may continue this process indefinitely, each time obtaining smaller and smaller positive integral values, the above equation is impossible. This method, called "infinite descent", was discovered by Fermat in the seventeenth century and is used now widely in various proofs.

Other mathematicians, such as R.D. Charnichael, have shown that  $x^4 - y^4 \neq *$ , in several different ways. To use his proof we must use the following lemmas:

Lemma I: There do not exist integers  $x, y, z$ , and  $t$ , all of which not equal to zero, such that  $t^2 + x^2 = 2z^2$  and  $t^2 - x^2 = 2y^2$ . (this can be proven by means of infinite descent, which is too lengthy to show here)

Lemma II: The area of a Pythagorean triangle is never equal to a square number. To sketch the proof of this lemma, let the legs and the hypotenuse of a

Pythagorean (right) triangle be  $u, v, w$  respectively. One may use, for this proof, a primitive right triangle since common factors may otherwise be divided out. Assume that  $u^2 + v^2 = w^2$  and  $uv = 2p^2$  is possible, where  $p$  is any positive integer and  $(u, v) = 1$ . Let  $u$  and  $v$  take on the values  $2ab$  and  $a^2 - b^2$  respectively. Hence  $p^2 = (ab)(a-b)(a+b)$ . Since  $a$  and  $b$  are taken to be relatively prime so as to satisfy  $(u, v) = 1$ ,  $ab$ ,  $a-b$ , and  $a+b$  are relatively prime to each other. Therefore  $a = m^2$ ,  $b = n^2$ ,  $a+b = r^2$  and  $a-b = s^2$ , or in other words  $m^2 - n^2 = s^2$  and  $m^2 + n^2 = r^2$ , which has been proven to be impossible by Lemma I. We are now able to prove that  $x^4 - y^4 = z^2$  is impossible.

Assume that  $x^4 - y^4 = z^2$ . Multiply both sides of this equation by  $x^2 y^2$  to obtain  $x^2 y^2 (x^4 - y^4) = z^2 x^2 y^2$ . Notice, however, that  $(2x^2 y^2)^2 + (x^4 - y^4)^2 = (x^4 + y^4)^2$ , or in other words, a right triangle with legs  $2x^2 y^2$  and  $x^4 - y^4$  has been formed. This triangle, however, has an area equal to  $z^2 x^2 y^2$  (a square number) and hence can not exist. Therefore  $x^4 - y^4 = z^2$  is impossible and  $x^4 + y^4 \neq z^4$  can be proven as a corollary.

As of yet, Diophantine analysts have failed in their search to find an invariant method by which a large class of problems, such as the finding of solutions of the general equation  $x^2 + ay^2 = z^n$ , might be completely solved; neither will I make that attempt. Instead, as a final illustration, I shall try only to systemize the discussion of the above equation. To do so, we need a fundamental lemma which I shall not prove.

Lemma: Two integers,  $t$  and  $u$ , and a multiplier can be bound in such a manner as to have  $t^2 + au^2 = \lambda z$ ,  $t \equiv 0 \pmod{z^n}$ , and  $|\lambda| \leq \sqrt{4/3}|a|$  with the greatest common divisor of  $t$  and  $u$  dividing  $\lambda$  and  $z$ , and  $N^2 \equiv -a \pmod{z^n}$ ,  $\lambda - Ny \equiv 0 \pmod{z^n}$ .

It is by the use of this lemma that one is able to prove that all solutions of  $x^2+ay^2=z^n$ , in which  $x$  and  $y$  are relatively prime and  $z=t^2+au^2$ , are gotten from the equation  $x+y\sqrt{-a}=(p+\sqrt{-a})(t+u\sqrt{-a})$ , in which  $t$  and  $u$  are relatively prime (as obtained from the first conditions), and are connected by the equation  $p^2+ae^2=(\pm 1)^n$ . This result may be found in most number theory books, but the margin here is too small to contain it.

All solutions of the equation  $x^2+y^2=z^3$ , where  $(x,y)=1$  and  $z>0$ , are given by  $(x+iy)=(p+iq)^3$ ,  $-1$  or  $1$ . Equating real coefficients yields  $x=p^3-3pq^2$  and  $y=3p^2q-q^3$ ; from this it is seen that  $z=p^2+q^2$ . This result could be gotten in a different way by seeing that  $(p^2+q^2)^3=(p^2+q^2)(p^2+q^2)^2=((p^2-q^2)^2+(2pq)^2)(p^2+q^2)=(p^3+pq^2)^2+(p^2q+q^3)^2$  (this is obtained by using the identity,  $(a^2+b^2)(c^2+d^2)=(ac-bd)^2+(ad+bc)^2$ ).

One may use this method to prove Fermat's statement that 25 is the only square to which when 2 is added will yield a cube. To show this, we proceed from the equation  $x^2+2=y^3$  to get  $x+\sqrt{-2}=(p+q\sqrt{-2})^3$ , whence, as before,  $x=p^3-6pq^2$ ,  $y=1-3p^2q-2q^3$ . Factoring the preceding equation yields  $q(3p^2-2q^2)=1$ . Consequently,  $q=\pm 1$ , from which  $3p^2-2=\pm 1$ . It is easily seen that  $3p^2-2=-1$  will not yield an integral solution for  $p$ , hence  $3p^2-2=+1$ . Therefore  $p=\pm 1$ ,  $q=1$ , from which we obtain the result that  $x=\pm 5$  are the only solutions of  $x^2+2=y^3$ . Fermat had gotten this result in 1657! As another exercise, the reader might like to solve  $x^2-2=y^3$ : (Hint: Set  $y=z-1$  and proceed in the above manner.)

Take  $x^2+y^2=z^n$  (letting  $a=1$ ), and  $N^2 \equiv -1 \pmod{z^n}$ , which can be proven impossible if  $z$  is even and  $n>2$ . (The proof is too complex to give here.) Let  $z$  be positive so that  $x=1$ ,  $t^2+u^2=z$ , and  $(t,u)=1$ . Then, since  $p^2+e^2=(\pm 1)^n$ , all solutions of  $(x+iy)=\epsilon(t+iu)^n$ , where  $\epsilon=1$  or  $i$ , are satisfied only by  $(p,\epsilon)=(1,0)$  or  $(0,1)$ . In particular, for  $n=2$  we get  $(x+iy)=(p+iq)^2$  or  $(x+iy)=i(p+iq)^2$ . By equating

real and imaginary parts of the first equation, we obtain  $x=p^2-q^2$ ,  $y=2pq$ , and likewise, from the second equation, we obtain  $x=2pq$ ,  $y=p^2-q^2$ . In these sets of equations,  $p$  and  $q$  must be relatively prime and of different parity in order for  $z$  to be odd. Consequently, the above parametric equations constitute the general integral solution of the equation  $x^2+y^2=z^2$ , where  $(x,y)=1$ .

There are many interesting problems in Diophantine analysis which demand special assumptions. One such problem is to solve the following system of equations in positive integers:  $x^2+y^2=a^2$ ,  $x^2+z^2=b^2$ , and  $y^2+z^2=c^2$ . The simplest solution is gotten by setting  $x^2=4mnpq$ ,  $y=mp-nq$ ,  $z=np-mq$ , so that  $x^2+y^2=(mp+nq)^2$  and  $x^2+z^2=(np-mq)^2$ . Since  $y^2+z^2=c^2$ , set  $y=2(m^2-n^2)rs$  and  $z=(m^2-n^2)(r^2-s^2)$ , so that  $y^2+z^2=(m^2-n^2)^2(r^2+s^2)^2$ . Now  $y=mp-nq=2rs(m^2-n^2)$  and  $z=np-mq=(r^2-s^2)(m^2-n^2)$ . Solving these two equations simultaneously for  $p$  and  $q$  yields  $p=2mrs-n(r^2-s^2)$ , and  $q=2nrs-m(r^2-s^2)$ . Now  $x^2=4mnpq=4mn(2mrs-n(r^2-s^2))(2nrs-m(r^2-s^2))$ , which when expanded gives a fourth degree equation in  $r$  and  $s$ . To make  $x^2/4$  a square, set  $r=4mn$  and  $s=m^2+n^2$ , leaving as a result the square of  $mnr^2-(m^2+n^2)rs+mns^2$ . (I have left the details to the avid reader.) We now obtain a two-parameter solution of  $(x,y,z)$ :  $x=2mn(3m^2-n^2)(3n^2-m^2)$ ,  $y=8mn(m^4-n^4)$ ,  $z=(m^2-n^2)(m^2-4mn+n^2)(m^2+4mn+n^2)$ . To obtain the smallest positive solution, we let  $m=2$  and  $n=1$ , obtaining the following solution (which the reader may verify):  $44^2+240^2=244^2$ ,  $44^2+117^2=125^2$ ,  $240^2+117^2=267^2$ .

WOCKYJABBER

-Hilbert Schenck, Jr.

And as in tedious thought he sat,  
The integral without a name,  
Rose from a skewed, conformal map,  
Diverging as it came!

'Twas finite and the polar cusp,  
Orthogonal to the secant lay.  
The semi-tacnode operates on  
The Gudermanian of A.

Pi-e, Pi-e, and x, y, z,  
The digital went clicky-clack.  
He found the norm in series form,  
And brought the work sheets back.

"Beware the integral, my son,  
With shape of non-symmetric bell,  
Beware old Van der Pol, and shun  
The curious vector del."

"Oh, hast thou solved the integral?  
Here is a raise, my brainish boy!"  
He threw his time cards in the air  
And clapped his hands with joy.

He took his program in his hand.  
Long hours the real root he sought.  
Then rested by the storage drums,  
And sat awhile in thought.

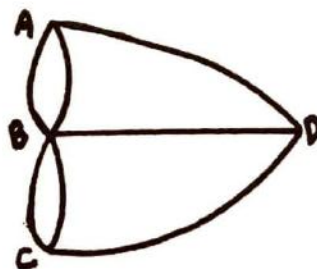
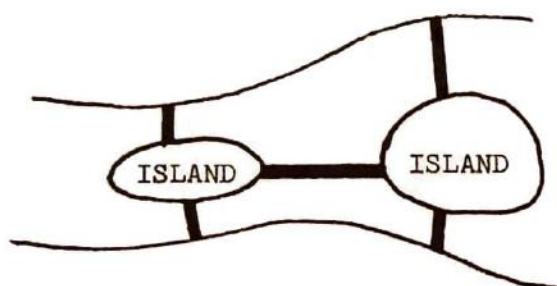
'Twas finite, and the polar cusp,  
Orthogonal to the secant lay.  
The semi-tacnode operates on  
The gudermanian of A.

# Topology

by Howard Elman

Once upon a time there was a town in Prussia called Koenigsberg. The people of this town had a favorite pastime. It seems that the town was divided into two parts by a river, and seven bridges crossed the river by way of two islands. The people, when taking their afternoon strolls, enjoyed trying to make an excursion across each bridge without crossing any bridge twice. They soon tired of this game, though, for repeated failure convinced them that it was impossible to win.

Leonhard Euler, one of the greatest mathematicians of all time, received word of this problem in his home in the court of Catherine the Great of Russia, and in 1735, he deciphered it. Euler simplified the problem by making each bridge a line, or arc, and each land mass a point, or vertex. The arcs are connected to the vertices in the same way that the bridges connect the land masses.



This is known as drawing the diagram as a graph (a drawing consisting of a finite number of vertices and a number of arcs). A vertex is defined as even if it has an even number of arcs extending from it, and odd if it has an odd number. The problem then becomes one of traversing the graph; that is, tracing each line of the graph in one continuous sweep of the pencil without lifting it from the paper or retracing any lines.

Euler discovered that if all the vertices of the graph are even, it may be traversed and it is possible to start and finish at the same point. If it has at most two odd vertices, it may be traversed, but it is not possible to start and finish at the same point. He deduced a general formula for the possibilities of traversing any graph, as follows: if the graph contains  $2n$  odd vertices (where  $n$  is any positive integer),  $n$  journeys are required to traverse it. Since all four vertices of the above graph are odd, it cannot be traversed in one journey, as the people of Koenigsberg suspected.

In solving this problem, Euler helped to found a whole new mathematical branch: topology. Topology is the branch of geometry dealing with the position or place of a figure. It is concerned with the geometrical properties that a figure retains if its size or shape is changed. From this definition it can be seen that the above problem is a topological problem; it has no concern with the lengths of the arcs or the sizes of the angles. No matter how these measures are altered, Euler's proof and formulas still hold.

Two objects are said to be topologically equivalent, or homeomorphic, if one object may be distorted in such a way as to look exactly like the other. (Such a transformation may take place if the objects are made of rubber that may be stretched or compressed in any way. Hence, topology has also been called rubber sheet geometry.) More specifically, in order for a topological transformation to take place between two objects,  $A$  and  $A'$  (that is, in order for them to be considered topologically equivalent), the two objects must live up to two conditions: (1) To each point  $p$  of  $A$  there corresponds only one point  $p'$  of  $A'$ , and conversely, to each point of  $A'$  there corresponds only one point of  $A$ . This is known as a biunique correspondence. (2) If two points  $p$  and  $q$  of  $A$  are moved such that the distance between them approaches zero,  $p'$  and  $q'$  of  $A'$  must also approach zero, and conversely. This condition provides that the points of homeomorphic figures are continuous in both directions.

It can thus be seen that two figures such as the circle and the triangle are topologically equivalent. If the circle is made of rubber, it may be stretched into the triangle. In doing this stretching, two precautions must be taken: the rubber must not be ripped (Ripping the material will violate condition two, because two points that would normally tend towards coincidence would be stopped by the split in the rubber.), and two points must not be brought into contact. (This would violate condition one, because two distinct points would become one.) It can also be seen that two other figures, such as the circle and the ring



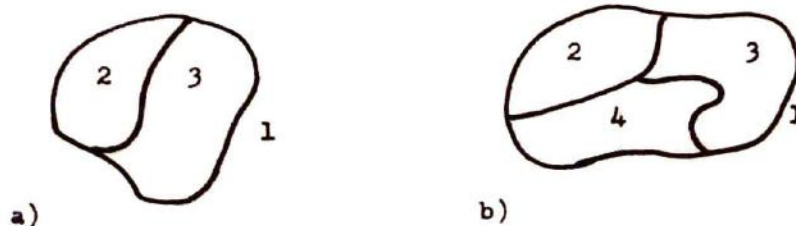
are not topologically equivalent. In order for the circle to become a ring, the rubber must be ripped. In order for the ring to become a circle, points must be brought into contact.

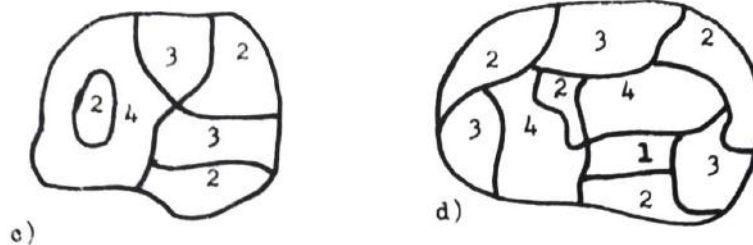
There are also topological classifications that apply only to three-dimensional objects. One is the genus of a surface. If a closed curve  $C$  is drawn on a sphere and the surface is cut along that curve, the cut will split the sphere into two parts, or to put it in another way, there are at least two points on the sphere which cannot be connected with a curve that does not intersect  $C$ . However, if a curve  $C'$  is drawn on a torus (a doughnut shaped object), the object will not split into two parts, and any two points on the torus may be connected with a curve that does not intersect  $C'$ . Two curves must be drawn to separate the torus into two distinct parts. This difference indicates that the sphere is topologically different from the torus. (Genus has no bearing on plane figures because any plane will be separated by one closed curve.)



A surface containing two holes requires three cuts to be separated. The genus of a surface has been defined as the largest number of nonintersecting closed curves that can be drawn on a figure without separating it. The genus of a sphere is zero; of a torus is one; etc. It has been proven that two surfaces of the same genus are topologically equivalent. A figure with  $n$  holes is topologically equivalent to a sphere with  $n$  handles.

A number of interesting problems have arisen in topology, the most notable being the four color problem. Consider the maps below:





All four represent islands that are controlled by different countries. The numbers represent colors used to identify different regions. Island (a) is divided between two countries and needs three colors: one for each country and one for the ocean. Island (b) is controlled by three countries, and needs four colors. However, island (c) is controlled by six countries and still needs four colors, as does island (d), which is controlled by ten countries. It seems that no more than four colors are needed for any map.

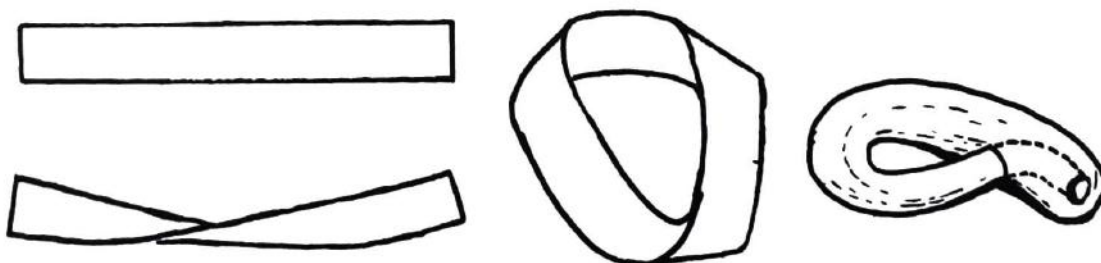
This theorem, though intuitively reasonable, has not been proven. It has been proven that five colors are sufficient for any map, and it has been shown that any map with less than thirty-eight regions requires only four colors, but no definitive proof has been devised for  $n$  regions.

Topology also concerns itself with the study of knots. The two knots below are almost identical; one is the mirror image of the other. They are topologi-



cally equivalent. However, one may not be transformed into the other, no matter how it is twisted.

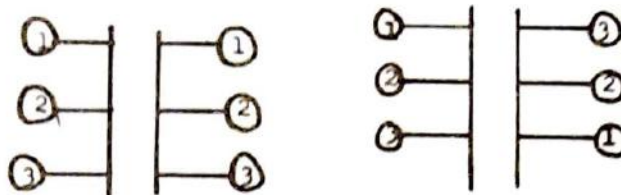
An unusual figure studied in topology is the Moebius strip. This is a surface with only one side. It can be made by taking a strip of paper, giving it a half twist, and then attaching the ends. If someone were to try to color one side of this strip, he would invariably cover the whole thing, since it has only one side. The Moebius strip has a number of other unusual properties. If the strip is cut along its center line, a surprising figure will be formed. If this figure is cut again, another surprise will take place. (The procedure is left to the reader.)



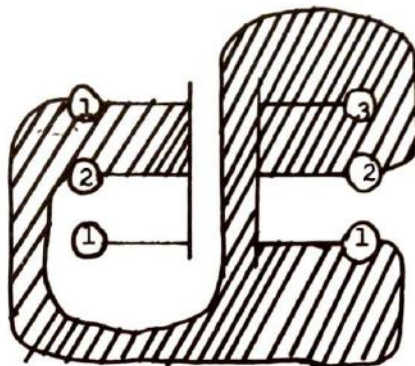
Another strange figure is the Klein bottle, which seems to have no inside or outside.

A topological theorem has led to the solution of an old problem. This theorem is the Jordan curve theorem, which states that any simple closed curve divides the plane into exactly two regions, an inside and an outside. Among other things, this statement provides justification for the use of Venn diagrams in any two-valued language.

This theorem also explains the problem of the Persian Caliph and his daughter. According to an old story, there was once a Persian Caliph who had a beautiful daughter. This Caliph was upset by the number of suitors that his daughter had, so he set up a contest, the winner of which would marry his daughter. The contest was made up of two parts. The first part consisted of connecting the corresponding numbers in the diagram at left with non-intersecting lines. All the participants did this with no trouble.



The second part consisted of connecting the corresponding numbers of the figure in the center. This part is not quite so easy. As a matter of fact, it is impossible. The reason is that when any two pairs of numbers are connected, a simple closed curve is formed, with one of the third numbers being inside and the other outside. According to the Jordan curve theorem, the numbers can't be connected without intersecting the curve. Unless the Caliph relented, it can be safely assumed that his daughter died an old maid.



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## MATH TEAM

The Math Team is an abelian group (all members commute) of dedicated?, aspiring?, mathematicians? who meet every day at 8:00 to discuss tricky problems in geometry, number theory, probability, advanced algebra, logic, lower arithmetic, and psychology. Quarterbacked by George Barany and Jay Banks, the team moved from forecasted mediocrity to reviling power in capturing a tie with Bronx Science in the fall Interscholastic Mathematics League. Douglas Fried led individual scorers in Stuyvesant (and the city, as well) with 27 out of a possible 30, followed by George Barany who scored 25. On the national Math Association of America (MAA) exam, Douglas Fried scored 127.5, George Barany 107.25, and Jay Banks 99.99999... for a team composite score of 334.75, which should rank among the top in the nation. Larry Auerbach also made the Honor Roll (80 or above) with a score of 91.

The variety of techniques used to attack problems is illustrated for the following example: "Find the smallest positive integral solution for the equation  $x + 2y = 3$ ."

1) Auerbach Iteration: Let  $f(x,y) = x + 2y - 3$

$$x_{n+1} = x_n - \frac{f(x_n, 0)}{\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}(x_n, 0)} \quad x_{n+1} = 2x_n + 3$$

Since this method appears to be unlucky, let  $x_3 = 13$ . Then  $x_2 = 5$ , so  $x=1, y=1$ .

2) Jacobs Literation (named after Jon Jacobs, junior member and top chess player on the team): The smallest integral is found by Euler's equation

$\frac{\partial I}{\partial y} = \frac{d}{dx} \frac{\partial I}{\partial y/dx} = 0$ . Implicit differentiation yields  $\frac{dy}{dx} = -\frac{1}{2}$ , so  $\frac{I}{y} = \infty$ ,  $\partial I = 1$  as  $y \rightarrow 0$ . The answer accordingly is  $x = 1, y = 1$ .

$\partial I = 1$  as  $y \neq 0$ . The answer accordingly is  $x = 1, y = 1$ .

3) Fried Finnaegling: Suppose at the point in question  $x = \lambda y$ . Then the characteristic linear equation is  $\lambda + 2 = 3, \lambda = 1, x = 1, y = 1$ .

4) Barany Bravado: Let us generate an ideal from the coefficients 1 and 2 (both of which are prime). This ideal is not principal, however we can set up an asymptotic approximation which shows that the minimum of  $x$  and  $y$  is less than  $10^{23}$ . By checking all of these cases, we find  $x=1, y=1$  works.

5) Banks Bluntness: The answer is obviously  $(x=1, y=1)$  ! Zzzzzzzzzzzzzz....

## OPTICAL ILLUSION

Which line is longer?

Answer: The one on the bottom.  
Didn't you learn anything from the circles?

The following is an actual math team competition which took place on December 4, 1970. The contestants are each allowed eleven minutes for each pair of problems. You may wish to try these under test conditions to see how you would perform. Outline of solutions given on page 28

1. If  $n$  is an integer,  $20 < n < 30$ , and  $n + n^3$  is represented by a five-digit decimal numeral with leftmost digit 1 and rightmost digit 8, find  $n$ .
2. Solve for real values of  $x$ :  $0 < x^2 - 4x < 5$
3. Given regular pentagon  $ABCDE$  with  $\overline{BE}$  intersecting  $\overline{AC}$  and  $\overline{AD}$  in points  $F$  and  $G$  respectively. The inch-length of  $\overline{FG}$  is  $(15 - 5\sqrt{5})/2$ . Find the inch-length of a side of the pentagon.
4. Find, correct to the nearest tenth, the value of 
$$\frac{\sqrt{10 + 2\sqrt{21}} + \sqrt{10 - 2\sqrt{21}}}{\sqrt{10 + 2\sqrt{21}} - \sqrt{10 - 2\sqrt{21}}}$$

5. The altitudes of a triangle measure 2, 3, and 4 inches respectively. Find the area of the triangle in square inches, and express your answer in the form  $a/\sqrt{b}$  where a and b are integers and b has no square integral factor except 1.
6. Find the prime factors of  $(x-1)(x-2)(x-3)(x-4) - 24$  in the coefficient domain of integers.

Cooper Square  
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Administrative  
Offices

May 1, 1971

To the Underclassmen of Stuyvesant High:

This is a neighborly salute to Stuyvesant High School and its interesting publication, Math Survey, and a reminder to its readers that Cooper Union has degree offerings of special interest to mathematics students.

Our freshman class entering in September, 1971, has already been selected and some of Stuyvesant's current seniors are enrolled. But for today's Stuyvesant juniors, sophomores, and freshmen there is still plenty of time to stroll over to nearby Cooper Union and explore the courses of study we offer beginning in September, 1972.

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Sincerely,



Herbert Liebeskind  
Director of Admissions

# Primitive Roots and the Theory of Indices

by Larry Auerbach

last year's Math Survey featured an excellent exposition on the subject of solving linear and quadratic congruences in the form of David Harbater's article, "Solving Congruences". The main concern of this article is the general binomial congruence  $x^n \equiv a \pmod{m}$ , and the extensive theory behind its solution. However, it might be a good idea to discuss congruences and residue systems before proceeding to our topic.

If the difference of two integers  $a$  and  $b$  is divisible by a third integer  $m$ , then we say  $a$  is congruent to  $b$  modulo  $m$  and write  $a \equiv b \pmod{m}$ . For example,  $36 \equiv 2 \pmod{17}$ ,  $14 \equiv 27 \pmod{13}$ . For a detailed discussion of the basic properties of congruences, along with other fundamental concepts of number theory the reader is referred to the Definition Section. If the reader is already familiar with the subject, he may skip over the aforementioned discussion without missing anything of great importance. The theorem on congruences in the Definition Section will henceforth be referred to as Thm. 1, and its various parts as Thms. 1a, 1b, etc.

It is fairly obvious that every integer is congruent modulo  $m$  to one and only one of the integers  $0, 1, 2, \dots, m-1$ . Suppose we partition the set of all integers into  $m$  distinct classes; all integers congruent to  $0$  modulo  $m$  will constitute one class, all integers congruent to  $1$  will constitute a second class, ..., all integers congruent to  $m-1$  will constitute an  $m$ -th class. We call each of these  $m$  infinite, disjoint classes a residue class modulo  $m$ ; the elements of these classes are called residues. If  $a \equiv b \pmod{m}$ , then we see that  $a$  and  $b$  belong to the same residue class modulo  $m$ , and hence we say  $a$  and  $b$  are residues of each other modulo  $m$ .

We call a set of  $m$  integers, each arbitrarily chosen from a different residue class modulo  $m$ , a complete residue system modulo  $m$ . Since no two elements are from the same residue class, then the elements of a complete residue system modulo  $m$  are incongruent pairwise modulo  $m$ . The most important complete residue system modulo  $m$  is the least residue system, consisting of the integers  $0, 1, \dots, m-1$ , each of which is called a least residue modulo  $m$ .

The subset of a complete residue system modulo  $m$  consisting of those residues relatively prime to  $m$  (see Def. Sec.) is called a reduced residue system modulo  $m$ , which we shall abbreviate  $R_m$ . If we were to exhibit an  $R_m$  consisting entirely of least residues, we would see that it consists of all the positive integers less than or equal to  $m$  and relatively prime to  $m$ .

We have the following theorem about residue systems;

Thm. 2: Let  $r_1, r_2, \dots, r_n$  be a complete, or a reduced, residue system modulo  $m$ , and let  $(a, m) = 1$ . Then  $ar_1, ar_2, \dots, ar_n$  is a complete, or a reduced, residue system, respectively, modulo  $m$ .

Proof: Suppose the theorem is false, i.e. that for  $i \neq j$ ,  $ar_i \equiv ar_j \pmod{m}$ . Then by Thm. 1e,  $r_i \equiv r_j \pmod{m}$ . Since no two members of the same residue system are congruent modulo  $m$ , the assumption that  $i \neq j$  is necessarily contradicted, and hence the theorem is proved.

We define the Euler phi-function  $\phi(m)$  as being equal to the number of residues in  $R_m$ . An equivalent definition would be that  $\phi(m)$  equals the number of positive integers less than or equal to  $m$  and relatively prime to  $m$ . For example,  $R_9 = 1, 2, 4, 5, 7, 8$  (least residue system), and so  $\phi(9) = 6$ . We have the following theorems about the  $\phi$  function:

Thm. 3: If  $p$  is a prime, then  $\phi(p) = p-1$ .

Thm. 4: If  $m = ab$  and  $(a, b) = 1$ , then  $\phi(m) = \phi(a)\phi(b)$ .

Thm. 5: If  $m = p_1^{a_1} \cdot p_2^{a_2} \cdots p_n^{a_n}$ , where the  $p$ 's are distinct primes and the  $a$ 's are positive integers, then  $\phi(m) = m(1-1/p_1)(1-1/p_2)\cdots(1-1/p_n)$ . For example,  $1500 = 2^2 \cdot 3 \cdot 5^3$ , hence  $\phi(1500) = 1500(1-1/2)(1-1/3)(1-1/5) = 400$ .

Thm. 6 (Euler's Theorem): If  $(a, m) = 1$ , then  $a^{\phi(m)} \equiv 1 \pmod{m}$ . If  $m$  is prime, then  $a^{m-1} \equiv 1 \pmod{m}$ , which is the famous corollary known as Fermat's Theorem.

Proof : Let  $r_1, r_2, \dots, r_{\phi(m)}$  be an  $R_m$ . Then by Thm. 2, the set  $ar_1, ar_2, \dots, ar_{\phi(m)}$  is an  $R_m$  also. Hence, corresponding to each  $r_1$ , there is a unique  $ar_j$  such that  $r_1 \equiv ar_j \pmod{m}$ . Furthermore, different  $r_1$  will have different corresponding  $ar_j$ . This means that the numbers  $ar_1, ar_2, \dots, ar_{\phi(m)}$  are just residues modulo  $m$  of  $r_1, r_2, \dots, r_{\phi(m)}$ , but not necessarily in the same order. Hence,

$$(ar_1)(ar_2)\dots(ar_{\phi(m)}) \equiv r_1 r_2 \dots r_{\phi(m)} \pmod{m}, \text{ which yields } a^{\phi(m)} (r_1 r_2 \dots r_{\phi(m)}) \equiv$$

$(r_1 r_2 \dots r_{\phi(m)}) \pmod{m}$ . By definition, all the  $r_i$  are relatively prime to  $m$ , and therefore their product is relatively prime to  $m$ . Hence, by Theorem 1e, we divide both sides of the congruence by the product  $r_1 r_2 \dots r_{\phi(m)}$  and get the resulting congruence  $a^{\phi(m)} \equiv 1 \pmod{m}$ .

There is nothing in Euler's Theorem which says that for  $(a, m) = 1$ , the congruence  $a^n \equiv 1 \pmod{m}$  is impossible for positive integers  $n < \phi(m)$ . However, there obviously must be a least positive exponent  $n$  satisfying the congruence. If  $h$  is that exponent, then we say  $a$  belongs to the exponent  $h$  modulo  $m$ . For example, 5 belongs to 4 modulo 13, 5 belongs to the exponent 256 modulo 257.

Thm. 7 : If  $a$  belongs to the exponent  $h$  modulo  $m$ , then  $h | \phi(m)$ . (The notation  $a | b$  means  $a$  divides  $b$ , and, analogously,  $a \nmid b$  means  $a$  does not divide  $b$ .) In the examples given in the previous paragraph, we see that  $4 | \phi(13) = 12$ ,  $256 | \phi(257) = 256$ . Proof: Suppose  $a$  belongs to  $h$  modulo  $m$  and  $h \nmid \phi(m)$ . Then  $\phi(m) = qh + r$ ,  $h > r \geq 1$ .

Then  $1 \equiv a^{qh+r} \equiv a^{qh} a^r \equiv (a^h)^q a^r \pmod{m}$ . Since  $h > r$ , this contradicts the hypothesis that  $a$  belongs to  $h$  modulo  $m$ , and hence the proof.

The following theorem will be applied later on;

Thm. 8: If  $a$  belongs to  $h$  modulo  $m$ , and  $(k, h) = d$ , then  $a^k$  belongs to  $h/d$  modulo  $m$ . Ex.- 5 belongs to 9 modulo 19,  $(6, 9) = 3$ , hence  $5^6$  belongs to  $9/3 = 3$  modulo 19.

Proof: Let  $a^k$  belong to the exponent  $n$  modulo  $m$ ; then  $(a^k)^n \equiv 1 \pmod{m}$ . By Thm. 7, we see that  $h | kn$ , which implies  $(h/d) | (k/d)n$ . Since  $(h/d, k/d) = 1$ ,  $h/d$  must divide  $n$ . Conversely,  $(a^k)^{h/d} = (a^h)^{k/d}$  and  $(a^h)^{k/d} \equiv 1 \pmod{m}$ ; hence, still by Thm. 7,  $n | (h/d)$ . But  $(h/d) | n$  as was shown, and therefore  $n$  must equal  $h/d$ .

Consider the series  $a, a^2, \dots, a^h \pmod{m}$ , where  $(a, m) = 1$  and  $a$  belongs to  $h$  modulo  $m$ . Each term of the series is relatively prime to  $m$  and therefore congruent modulo  $m$  to a member of  $R_m$ . Also, no two terms are congruent modulo  $m$ ;

this is easy to show. Suppose, for  $h > r > s \geq 1$ ,  $a^r \equiv a^s \pmod{m}$ . Then  $a^{r-s} \equiv 1 \pmod{m}$ , but, since  $r-s < h$ , our assumption that  $a$  belongs to  $h$  modulo  $m$  is contradicted, and hence our proof. From this, we can conclude that the series forms a subset of  $R_m$  consisting of  $h$  distinct residues.

Now, if  $h = \phi(m)$ , the series  $a, a^2, \dots, a^h \pmod{m}$  forms an  $R_m$ , since it forms a subset of  $R_m$  with  $\phi(m)$  distinct terms. We have a special name for an  $a$  belonging to  $\phi(m)$  modulo  $m$ ; we call it a primitive root modulo  $m$ . For example, 2, 6, 7, and 11 are all primitive roots modulo 13. Thm. 8 allows us to make some interesting conclusions about primitive roots, as shown below.

Thm. 9 : If  $g$  is a primitive root modulo  $m$ , then  $g^k$  is one also, if and only if  $(k, \phi(m)) = 1$ . The proof follows if we let  $h = \phi(m)$  in Thm. 8.

Thm. 10 : If  $m$  has any primitive roots, then it has exactly  $\phi(\phi(m))$  distinct ones.

Proof : If  $g$  is a primitive root modulo  $m$ , then  $g, g^2, \dots, g^{\phi(m)}$  forms an  $R_m$ , and any primitive root modulo  $m$  must occur among them. Let  $\phi(m) \geq k \geq 1$ . By Thm. 8,

$g^k$  belongs to the exponent  $n = \phi(m)/(k, \phi(m))$ . Now,  $g^k$  is a primitive root modulo  $m$  precisely when  $n = \phi(m)$ ; that is, when  $(k, \phi(m)) = 1$ . The number of integers  $k$  satisfying  $\phi(m) \geq k \geq 1$  and relatively prime to  $\phi(m)$  is, by definition,  $\phi(\phi(m))$ , and hence the proof.

In the statement of Thm. 10, we notice the phrase "If  $m$  has any primitive roots, ...". The implication is obvious; there are moduli for which no primitive roots exist. In fact, the only moduli possessing primitive roots are

2, 4,  $p^n$ , and  $2p^n$ , where  $p$  is an odd prime and  $n$  is a positive integer. Although a proof of this statement is elementary, it is rather lengthy. Interested readers can find the proof in any good number theory book.

Suppose  $g^n \equiv a \pmod{m}$ , where  $g$  is a primitive root modulo  $m$  (thus imposing restrictions on  $m$  as mentioned), and where  $n$  is a positive integer. Then we say  $n$  is the index of  $a$  modulo  $m$  to base  $g$ , denoted  $\text{ind}_g a$  (more precisely  $\text{ind}_g a$ , but the base is usually omitted). We see that  $\text{ind}_g a$  also equals  $n + k\phi(m)$  for some non-negative integer  $k$ , since  $g^{n+k\phi(m)} \equiv g^n g^{k\phi(m)} \equiv g^n \pmod{m}$ . From this, we conclude that indices of  $a$  to base  $g$  are congruent modulo  $\phi(m)$ , and hence all index arithmetic is performed modulo  $\phi(m)$ . We have the following theorem about the properties of indices:

Thm. 11 : Let  $a$  and  $b$  be in  $R_m$ . Then:

- If  $a \equiv b \pmod{m}$ , then  $\text{ind}_g a \equiv \text{ind}_g b \pmod{\phi(m)}$ .
- $\text{ind}_g ab \equiv \text{ind}_g a + \text{ind}_g b \pmod{\phi(m)}$ .
- $\text{ind}_g a^n \equiv n \cdot \text{ind}_g a \pmod{\phi(m)}$ .
- If  $g$  and  $q$  are primitive roots modulo  $m$ , then  $\text{ind}_g q \cdot \text{ind}_q a \equiv \text{ind}_g a \pmod{\phi(m)}$ .
- $\text{ind}_g 1 \equiv 0 \pmod{\phi(m)}$ .
- $\text{ind}_g g \equiv 1 \pmod{\phi(m)}$ .

The analogy between indices and logarithms is almost too apparent.

Just as one needs a log table to do logarithm arithmetic, one needs an index table to do index arithmetic. Obviously, different moduli require different tables. Let us construct a sample table for, say, modulo 13. We must choose a base (primitive root) - any  $g$  will do. So, let  $g = 2$  (usually the least positive primitive root is chosen for table construction).

| $a \pmod{13}$    | 2 | 4 | 8 | 3 | 6 | 12 | 11 | 9 | 5 | 10 | 7  | 1  |
|------------------|---|---|---|---|---|----|----|---|---|----|----|----|
| $\text{ind}_2 a$ | 1 | 2 | 3 | 4 | 5 | 6  | 7  | 8 | 9 | 10 | 11 | 12 |

Thm. 9 enables us to find the other primitive roots modulo 13. We must find those  $\text{ind}_2 a$  relatively prime to  $\phi(13) = 12$ ; their "anti-indices" would be the other primitive roots. So, those  $\text{ind}_2 a$ , excluding 1, are 5, 7, and 11; hence, the other primitive roots modulo 13, besides 2, are 6, 7, and 11. Theorem 10 is verified, since  $\phi(\phi(13)) = \phi(12) = 4$ , and there are four primitive roots. (From now on, we will use the term anti-ind  $a$  to mean the number whose index is  $a$ .)

We can now begin our conquest of the congruence  $x^n \equiv a \pmod{m}$ , which, of course, was the objective of this article. Taking indices, the linear congruence  $n \cdot \text{ind}_g x \equiv \text{ind}_g a \pmod{\phi(m)}$  results. Hence, all we need do is solve for  $\text{ind}_g x$ ; however, before doing so, we must first learn a few things about linear congruences.

Thm. 12: If  $(a, m) = 1$ , then  $ax \equiv b \pmod{m}$  has the unique solution  $x \equiv ba^{\phi(m)-1} \pmod{m}$ . Proof: Consider  $ax \equiv b \pmod{m}$  as  $x$  ranges from 0 to  $m-1$ . Then, by Thm. 2, the set  $0, a, 2a, \dots, (m-1)a$  forms a complete residue system modulo  $m$ , since  $0, 1, 2, \dots, m-1$  forms one. Therefore  $b$  corresponds to a unique element of the system, and hence a unique solution for  $x$ . Now, by Thm. 6, we have the congruence

$a^{\phi(m)} \equiv 1 \pmod{m}$ , and so  $a^{\phi(m)-1}(ax) \equiv a^{\phi(m)} x \equiv ba^{\phi(m)-1} \pmod{m}$ , which is the unique solution for  $x$ .

Thm. 13 (generalization of Thm. 12): The congruence  $ax \equiv b \pmod{m}$  has no solutions or exactly  $d = (a, m)$  solutions, depending upon  $b$  is not, or is, congruent to 0 modulo  $d$ .

Proof: When  $ax \equiv b \pmod{m}$  has a solution for  $x$ , it is clear that  $d = (a, m)$  must be a divisor of  $b$ . If we see that is so, then we see that the congruence

$(a/d)x \equiv (b/d) \pmod{m/d}$  has the unique solution  $x_0 \equiv (b/d)(a/d)^{\phi(m/d)-1} \pmod{m/d}$  by Thm. 12. Then  $ax \equiv b \pmod{m}$  is satisfied by all the numbers  $x_0 + n(m/d)$  for an arbitrary integer  $n$ , and only by these numbers. Hence, the incongruent solutions of  $ax \equiv b \pmod{m}$  may be represented by the  $d$  integers  $x_0 + n(m/d)$  for  $n = 0, 1, \dots, d-1$ .

Now, we have  $n \cdot \text{ind } x \equiv \text{ind } a \pmod{\phi(m)}$ . Let  $(n, \phi(m)) = d$ . Then, by Thm. 13, there is a solution for  $x$  if and only if  $\text{ind } a \equiv 0 \pmod{d}$ . Suppose there is a solution. Then  $(n/d) \cdot \text{ind } x \equiv (\text{ind } a)/d \pmod{\phi(m)/d}$ , and by Thm. 12, we have  $\text{ind } x \equiv (n/d) \phi(m/d)^{-1} (\text{ind } a)/d \pmod{\phi(m)/d}$  (whew!). Converting back to modulo  $\phi(m)$ , we have the  $d$  solutions for  $\text{ind } x$ . We then take their anti-inds, and we have the  $d$  solutions for  $x$ .

To better illustrate the method of solution, let us solve three sample problems, all utilizing the modulo 13 index table.

- 1)  $x^9 \equiv 11 \pmod{13}$ . Then  $9 \cdot \text{ind } x \equiv \text{ind } 11 \pmod{12}$ , and  $(12, 9) = 3$ . But  $\text{ind } 11 = 7$ , and  $7 \not\equiv 0 \pmod{3}$ . Hence, there is no solution.
- 2)  $x^5 \equiv 9 \pmod{13}$ . Then  $5 \cdot \text{ind } x \equiv \text{ind } 9 \pmod{12}$ , and  $(5, 12) = 1$ . By Thm. 12, there is a unique solution. The actual computation of the solution goes as follows.  $5 \cdot \text{ind } x \equiv 8 \pmod{12}$ , and  $\phi(12) = 4$ . Then  $\text{ind } x \equiv 8 \cdot 5^{-1} \equiv 1000 \equiv 4 \pmod{12}$ . anti-ind 4 = 3, and hence  $x \equiv 3 \pmod{13}$ .
- 3)  $x^8 \equiv 3 \pmod{13}$ . Then  $8 \cdot \text{ind } x \equiv \text{ind } 3 \pmod{12}$ ,  $(12, 8) = 4$ ,  $\text{ind } 3 = 4$ , and  $4 \equiv 0 \pmod{4}$ ; hence, there are four distinct solutions for  $x$ .  $8 \cdot \text{ind } x \equiv 4 \pmod{12}$ , so  $2 \cdot \text{ind } x \equiv 1 \pmod{3}$ , since we can divide 8, 4, and 12 by 4. Then  $\phi(3) = 2$ , so  $\text{ind } x \equiv 1 \cdot 2^{-1} \equiv 2 \pmod{3}$ . Therefore, the four solutions for  $\text{ind } x$  would be  $2 + (12/4)n$  for  $n = 0, 1, 2, 3$  (since  $d = 4$  and  $d-1 = 3$ ); more explicitly,  $\text{ind } x = 2, 5, 8$ , and  $11$ . The anti-inds of 2, 5, 8, and 11 are 4, 6, 9, and 7 respectively, which are the solutions for  $x$  modulo 13.

Similarly, one can solve exponential congruences like  $7^x \equiv 8 \pmod{41}$  using indices; the reader may wish to develop a method for doing so.

There are many related topics in number theory worthy of investigation by the reader. Some examples would be : vector indices and solving  $x^n \equiv a \pmod{m}$  when  $m$  has no primitive roots, indices modulo  $2^n$  ( $n$  greater than 2), methods for finding primitive roots (indispensible in the construction of index tables), primitive  $\lambda$  (lambda) roots, periodic decimals, primitive roots and their applications in group theory, the quadratic and higher reciprocity laws, ....



#### OPTICAL ILLUSION

④

⑧

Although the number on the left seems bigger, the number on the right is actually TWICE as large!

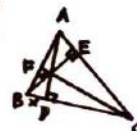
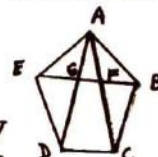
## Problems

- 1- Find the times between 7 and 8 o'clock when the hands of a clock will form an angle of 84 degrees.
- 2- Express in simplest form:  $[2^{n+4} - 2(2^n)]/2(2^{n+1})$
- 3- Point  $P(p,q)$  is a variable point on the parabola  $y=x^2$ .  $V(0,0)$  is the vertex of the parabola.  $R(0,r)$  is the point where the perpendicular bisector of  $VP$  intersects the y-axis. Find the limit of  $R$  as  $p$  approaches 0.
- 4- Simplify for  $x$ :  

$$x = \sqrt{1+\sqrt{1+\sqrt{1+\dots}}}$$
- 5- An eccentric millionaire wished to distribute a million dollars in gifts of either one dollar or powers of 7 such as 7; 49; 343; 2401, etc. He had an aversion to giving more than 6 people the same sum of money, but no restriction on the number of people who might taste of his bounty. How did he did he distribute the million dollars?
- 6- Find the smallest and largest squares in each of which all 10 digits are used only once.
- 7- A student accidentally erased a long-division example. He could recall only that the successive subtrahends starting from the top were 690; 2415; and 2070 and that the remainder was 1. With this information he restored the problem. Can you do likewise?
- 8- Prove that the product of 3 consecutive numbers is exactly divisible by 504 if the middle one is a cube.
- 9- "Find me a number," said the king to his jester, "half of which is a square." "How simple," said the jester. "But," continued the king, "one-third of it must be a cube." The jester looked serious for once. "And finally," remarked the king, "one-fifth of it must be a fifth power." The jester looked crestfallen. "Solution by tomorrow morning," finished the king, "or you must marry my widowed mother-in-law." The jester was crushed. But the next morning he had the answer and he was saved from a fate worse than death. See if you can find the number.
- 10- A tramp rolls cigarettes from the butts he picks up on the street. He finds that four butts make one new cigarette. How many cigarettes can he smoke from a haul of sixteen butts?
- 11- A tramp picks up apples off the corner applecart while the vender isn't looking. He finds he can steal 4 apples in an hour. In how many hours can he steal 16 apples?
- 12- I drive at an average speed of 30 miles per hour to the railroad station each morning and just catch my train. On a particular morning there was a lot of traffic and at the halfway point I found I had averaged only 15 miles per hour. How fast must I drive for the rest of the way to catch my train?

.....  
 Solutions for Math Team Problems found on page.22

1. By testing all of the numbers from 0 to 9, it is easily seen that in order for  $n + n^3$  to end in an eight,  $n$  must end with a 4 or 9.  $24 + (24)^3 = 13848$ , and  $29 + (29)^3 > 20,000$ . Therefore  $n = 24$ .
2. Solve the two inequalities,  $0 < x^2 - 4x$  and  $x^2 - 4x < 5$ , separately. The intersection of the two solution sets is the required answer. ( $4 < x < 5$  or  $-1 < x < 0$ )
3. In the accompanying diagram, let  $FG=x$  and  $FB=y$ . It then can be shown that  $EDFC$  is a rhombus with sides of length  $x+y$ . By similar triangles,  $FG/DC = AF/AC$ , or  $x/x+y = y/x+2y$ . Solve for  $y$  in terms of  $x$ , substitute for  $x$   $(15-5\sqrt{5})/2$ , and finally find the value of  $x+y$ . Answer is 5.
4. Multiply the numerator and denominator of the fraction by the conjugate of the denominator and simplify. The answer is 1.5.
5. In the accompanying figure,  $AD=2$ ,  $BE=3$ , and  $CF=4$ . Let  $BC=6a$  (thus making the area of  $ABC$  be equal to  $6a$ ), thus,  $AB=3a$  and  $AC=4a$ . Now let  $BD=x$ . By the Pythagorean Theorem,  $x^2 + 4 = 9a^2$  and  $(6a-x)^2 + 4 = 16a^2$ . Solve the two equations simultaneously to find  $a^2$ . Now find  $6a$  (the area of triangle  $ABC$ ). the answer is  $144/\sqrt{455}$ .
6. Expand the polynomial by multiplying the outer pairs and the inner pairs separately to obtain  $(x^2-5x+6)(x^2-5x+4) = 24$ . Let  $b=x^2-5x+4$ . Thus  $(b+2)b = 24$  is the equivalent polynomial. Solve this quadratic for  $b$  and resubstitute  $b=x^2-5x+4$ . Factor the terms which remain. Answer--  $x(x-5)(x^2-5x+10)$



# Compilers

by Seth Steinberg

Many people are aware of the rapid encroachment of computers on our lives and worry although computer experts inform us that computers are relatively stupid. Computers are intrinsically stupid and extremely docile and obey orders well. Unfortunately because of this one has to learn their language, a series of bits.

The use of computer languages such as FORTRAN requires the writing of a long program called a compiler. This enables the programmer to write in relatively normal looking instructions and get the results he expects.

FORTRAN stands for FORMula TRANslation. The ability of the compiler to handle the formulas or arithmetic expressions is its strongest point. The operations addition, subtraction, multiplication, division, and exponentiation are represented by +, -, \*, /, \*\* (a double star). A typical statement could be:

$A+B \times C \times D - E \times F / G$ .

Another feature is the use of functions. In FORTRAN one can write SIN(X) for the sine of x or EXP(-X\*\*2/2.0) for  $e^{-x^2/2}$ . Similar notation is used for representing an element of a vector or tensor. So  $a_{ij}$  is written A(I,J) and so on.

This article discusses a compiler of a language much simpler than FORTRAN, but uses basic techniques used in most compilers.

The symbols for addition, subtraction, multiplication, division, and exponentiation are +, -, \*, /, \*\*. All variable names are one letter long, which allows 26 names. The functions are S(X) sin x, C(X) cos x, A(X) tan<sup>-1</sup>x, H(X) tanh x, B(X) x, E(X)  $e^x$ , Q(X)  $\sqrt{x}$ , I(X) [x], and D(X) the sign of x. The other letters can be the names of vectors, which must be defined before the arithmetic statement references them.

So valid statements could be:

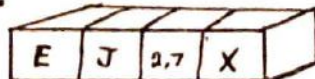
$(E(X) - E(-X)) / 2$

$S(X)^2 + C(X)^2$

$D/R \times T(J) - D/T + (G-K) \times M(K) - 3.14 \times M + 5$ .

The arithmetic processor has 3 states in its scan of the expression. The first looks for a unary operation, a(, or a name. The second accepts a( or a name. The last only accepts operators or parentheses. A special section of the compiler evaluates the numbers.

All the data used must have a location in core associated with it. Vectors require several locations.



The compiler has several tables to store data. One table holds symbols and their core allocations.

A 100

G 110

N 107

Another table holds numbers and the locations they will be put in.

23.9 106

897.58 117

Another holds the names of vectors and functions. Functions are indicated by a zero length and the machine code to perform them is in the other location. Vectors have a positive length and a core location.

| symbol | location or code     | length or zero |
|--------|----------------------|----------------|
| S      | the instruction SINE | 0              |
| X      | 195                  | 50             |

In the article the following machine codes are used:

LOAD value in core into accumulator

STOR puts accumulator into core storage

ADD adds a value to the accumulator

SUB subtracts a value from the acc.

MPY multiplies the accumulator by a value

DIV divides the acc. by a value

EXPN raises the acc. to a power

CSGN changes the sign of the acc.

SUBS fetches an element of a vector and the operators for the various functions.

The compiler goes from left to right and starts in state one with the level of parentheses equal to zero. It uses a table to determine how it handles the data.

|       |    | the data/ a number a letter + - / ( ) |   |   |   |   |   |    |   |   |  |
|-------|----|---------------------------------------|---|---|---|---|---|----|---|---|--|
| the   | 1/ | 1                                     | 2 | 3 | 4 | 0 | 0 | 0  | 5 | 0 |  |
| state | 2/ | 1                                     | 2 | 0 | 0 | 0 | 0 | 0  | 5 | 0 |  |
|       | 3/ | 0                                     | 0 | 8 | 8 | 9 | 9 | 10 | 7 | 6 |  |

The number indicates an operation to be performed:

- 0 generate an error message
- 1 get the value of the number and assign a core location for it, put both in the table, put the core location in the special string
- 2 look this up in the table of names, assign a location if it wasn't in the table, put the table and the location in the table, and the special string
- 3 ignore it
- 4 put 3 plus the level of parentheses into the special string
- 5 add 6 to the level
- 6 subtract 6 from the level
- 7 set the last location in the special string to zero, put the level plus 5 add 6 to the level of parentheses
- 8 put the level plus one
- 9 put the level plus two
- 10 put the level plus 4

After operations 5 or 7 the state becomes 1, after 3,4,8,9,10 it becomes 2, after 1,2, or 6 it becomes 3, after a 0 the compiler can stop since it will not be able to process it.

For example:

the string: S ( X ) \* ( - 3 2 . 5 + 8 ' M )  
 special : 0 5 -102 0 2 0 9 0 0 0 -107 7 -111 10 -98 0  
 string

The final level must be zero or the parentheses do not balance. This ends what is called the lexical scan.

Now the compiler looks for the highest number in the string just generated. If it is less than zero or equal to zero the compilation is complete. Otherwise it takes the number found modulo six. In this example the highest number is 10 modulo 6 is 4. This means the operation is an exponentiation. The compiler looks for the first negative numbers to the left and right of the operation. The code generated is:

LOAD 111  
 EXPN 98

The compiler sets the -107 and -98 to zero. A temporary address is taken from a list kept by the compiler and replaces the 10. A temporary address is a location in core used to store the intermediate results of an operation. The temporary address is also put in the accumulator indicator. This tells that the accumulator has useful data in it and gives a location to put it in. The special string becomes:

0 5 -102 0 2 0 9 0 0 0 -107 7 0 -110 0 0  
 and still corresponds to the string of characters.

The next operation is a 9 modulo 6 = 3. The unary minus has only one argument on the right. The required code is:

CSGN

However the accumulator indicator is set. If the item to load i.e. 107 is equal to the indicator then the LOAD is skipped and no new temporary address is generated. Otherwise the compiler stores the accumulator and loads the argument. Since the old value of the accumulator is saved at the temporary location, the new value requires a new location. In all cases the arguments are set to zero and the indicator replaces the operation code.

Since addition and subtraction are both one and division and multiplication are both two, the compiler looks at the character in the original string to determine the operation. These operations are handled like exponentiation except that the appropriate operation is generated.

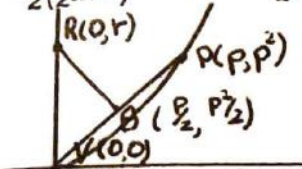
The functions or subscripted arrays are all 5. To find the correct operation the compiler first takes the character immediately before the one which corresponds to the 5. This is looked up in the table of functions. If the length is zero then it is a function whose code is in the location column. If this code is zero then an error results. If the length is not zero then the instruction SUBS followed by the argument, the location and then the length. This instruction puts the appropriate value in the accumulator.

This instruction group is handled like the unary minus as far as getting, LOADING, and replacing arguments goes. A temporary address is generated if necessary.

After the compilation of the arithmetic expression is complete the code generated is put in the accumulator unless an error occurred.

A language can be built around this processor by using keywords and ingenuity. Compiler programs tend to become large and hard to follow and some can take a long time to write. Once the arithmetic processing portion is written it can be used to develop useful languages by changing the instructions used. Happy compiler writing.

## Solutions

- 1)  $84^\circ$  is  $84/360$  of 1 hour.  $(60)(84/360) = 14$  minutes.  $14+x = 35+x/12$   
 $\therefore 11x/12 = 49$  or  $21$ ;  $x = 588/11$  or  $252/11$ ; thus,  $x = 53 \frac{5}{11}$  or  $x = 22 \frac{10}{11}$ .
- 2)  $\frac{2^{n+4} - 2(2^n)}{2(2^{n+1})} = \frac{2^{n+4} - 2^{n+1}}{2^{n+2}} = 2^2 - 2^{-1} = 4 - 1/2 = 7/2$
- 3) 

The equation of the line PV is  $y = px$   
 The equation of the normal RB is  $y = -x/p + r$   
 The two lines intersect at  $x = p/2$ , i.e.  $p(p/2) = p^2/2$   
 $= (-1/p)(p/2) + r = -1/2 + r$   
 $r = p^2/2 + 1/2$   $\lim_{p \rightarrow 0} r = \lim_{p \rightarrow 0} p^2/2 + 1/2 = 1/2$
- 4)  $x = \sqrt{1+\sqrt{1+\dots}}$   $x^2 = 1 + \sqrt{1+\sqrt{1+\dots}}$  By the quadratic formula,  $x = (1+\sqrt{5})/2$   
 Since  $x$  is positive,  $x = (1+\sqrt{5})/2$
- 5) Writing 1,000,000 in base 7, we have:  
 $(1,000,000)_{10} = (11333311)_7$  so that  
 $1,000,000 = 7^7 + 7^6 + 3 \cdot 7^5 + 3 \cdot 7^4 + 3 \cdot 7^3 + 7^2 + 7 + 1$   
 $(1,000,000)_{10} = (11333311)_7$  so that  
 $1,000,000 = 7^7 + 7^6 + 3 \cdot 7^5 + 3 \cdot 7^4 + 3 \cdot 7^3 + 7^2 + 7 + 1$   
 The millionaire, therefore, distributed his money as follows:  
 One gift each of  $7^7 = \$823,543$ ,  $7^6 = \$117,649$ ,  $7^5 = \$16,807$ ,  $7^4 = \$2,401$ ,  $7^3 = \$243$ ,  $7^2 = \$49$ ,  
 and three gifts each of  $7^5 = \$16,807$ ,  $7^4 = \$2,401$ ,  $7^3 = \$243$ ,  $7^2 = \$49$ .
- 6) There are altogether 87 squares using all 10 digits. The smallest is  $32043^2 = 1026753849$ . The largest is  $99066^2 = 9814072356$ .
- 7) Since each of the subtrahends is divisible by the divisor  $d$  in the example, and each must be less than  $10d$ ,  $d$  must equal 345 (the g.c.d. of the subtrahends).  $690 = 2 \cdot 345$ ,  $2415 = 7 \cdot 345$ ,  $2070 = 6 \cdot 345$ , and there is a remainder of 1, hence the dividend must equal  $(276)(345) + 1 = 95221$ .
- 8)  $y = (x^3 + 1)(x^3)(x^3 - 1)$ . By Euler's Theorem,  $(x^3 + 1)(x^3 - 1) = x^6 - 1$  is divisible by 7 and 9, since  $\phi(9) = \phi(7) = 6$ . Since  $(7,9) = 1$ ,  $y$  is divisible by  $7 \cdot 9 = 63$ . If  $x$  is even, then 8 divides  $x^3$ , and so  $8 \cdot 63 = 504$  divides  $y$ . If  $x$  is odd, then one of  $x^3 + 1$  and  $x^3 - 1$  is divisible by 4 and the other by 2. Therefore 8 divides  $(x^3 + 1)(x^3 - 1)$ , and hence  $(x^3 + 1)(x^3)(x^3 - 1)$  is divisible by  $8 \cdot 63 = 504$ .
- 9) A number  $n$ , divisible by 2, 3, and 5, may have the form  $n = 2^a 3^b 5^c$ . Since  $n/2$  is a square,  $a$  must be odd and  $b$  and  $c$  even. Similarly,  $a$  and  $c$  must be multiples of 3 and  $b \equiv 1 \pmod{3}$ . Also,  $a$  and  $b$  must be multiples of 5 and  $c \equiv 1 \pmod{5}$ . The smallest values satisfying these conditions are  $a = 15$ ,  $b = 10$ ,  $c = 6$ , and  $n = 2^{15} 3^{10} 5^6 = 30,233,488,000,000$ .
- 10) A butt is left over after the  $n$ -th cigarette is consumed, which is then used in the  $(n+1)$ th cigarette. Since one butt always remains, the same butt can be used in all cigarettes, with the remaining 15 of the 16 butts open for consumption. 3 new butts are used for each new cigarette, and hence 5 cigarettes can be manufactured by the tramp.
- 11) This problem is beyond the abilities of any of the staff members.
- 12) Suppose it takes  $t$  hours at 30 mph to reach the station. Therefore, it takes  $2t$  hours at 15 mph to do the same. Since  $1/2$  the distance has been traveled,  $t$  hours have passed, hence if the driver is to reach the station in time, he must cover 15t miles in 0 seconds, which is difficult.

Complete solution for geometry problem on inside cover:

Draw  $\overline{BZ}$  so that  $\angle ZBC = 20^\circ$ . Join  $\overline{XZ}$ . It is easily seen that  $\angle BZC = 80^\circ = \angle BXC$ . Thus,  $\overline{BZ} = \overline{BC}$ . In the same manner,  $\angle BXC = 50^\circ = \angle BCX$ . Thus,  $\overline{BX} = \overline{BC}$ . Now, since  $\angle XBZ = 60^\circ$  and  $\overline{BX} = \overline{BZ}$  (by transitivity),  $\triangle BXC$  is equilateral, and thus  $\overline{XZ} = \overline{BZ}$ .  $\angle BYZ = 180^\circ - \angle YBC - \angle YCB = 40^\circ = \angle YBZ$ . Consequently,  $\overline{YZ} = \overline{BZ}$ .

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